

Confidence & Uncertainty Estimation Methods on LLMs	Disadvantages
Input: What's the capital of France? LLM Output It is Paris Prob. 0.8 0.9 0.6 $Norm(\cdot)$ Conf. 0.75 ① Probability-based method (<i>Prob.</i>)	a. Requires normalization due to variable sequence length; b. Requires access to token-level probabilities, inapplicable to black-box LLMs; c. Fails to capture semantic meaning over token-level probabilities.
Input: What's the capital of France? LLM Output: It is Paris ⊕ Prompt: Is your answer True? LLM True 0.9 Conf. 0.9 ② $p(\text{True})$-based method ($p(\text{True})$)	a. Relies on prompting strategies to elicit confidence estimation, varying in different prompts; b. Cannot improve LLM's intrinsic confidence estimation ability. c. Requires access to token-level probabilities, inapplicable to black-box LLMs; d. Prone to be over-confident.
Input: What's the capital of France? LLM Output: It is Paris ⊕ Prompt: Is your answer True? LLM 0.85 Conf. 0.85 ③ Self-verbalized method (<i>Verb.</i>)	a. Relies on prompting strategies to elicit confidence estimation, varying in different prompts; b. Cannot improve LLM's intrinsic confidence estimation ability. c. Prone to be over-confident.
Input: What's the capital of France? LLM Output: It is Paris ✓ Output: It is Berlin ✗ Output: Paris. ✓ $Agg(\cdot)$ Conf. 0.66 ④ Sampling-based method	a. Requires additional inference time cost; b. Varying in different aggregation methods; c. Cannot improve LLM's intrinsic confidence estimation ability.
Input: What's the capital of France? LLM Output: It is Paris Evaluator Conf. 0.72 Training ⑤ Training-based method	a. Requires training an additional evaluator; b. Difficult to learn LLM's intrinsic confidence estimation on unseen domains.